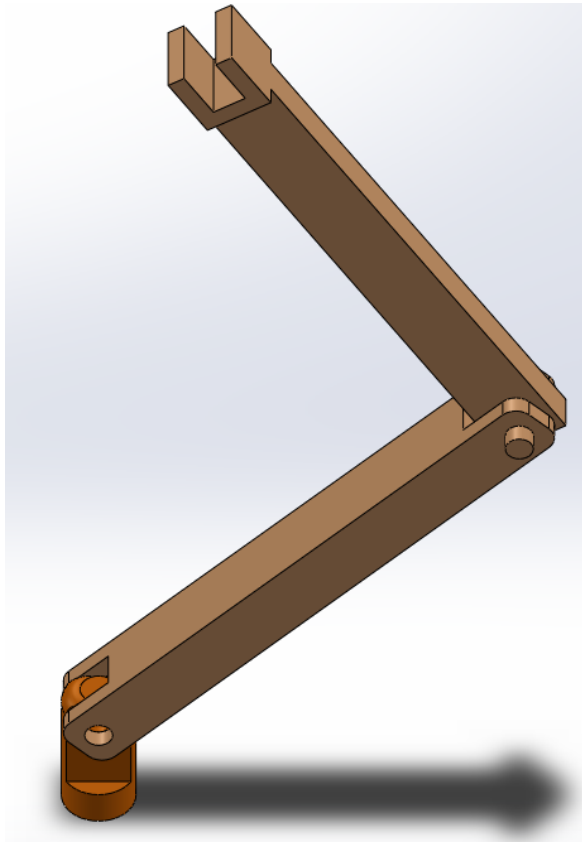


Kinematics of 3R or RRR Planar Robotic Arm

Presented by: Sourav Kumar

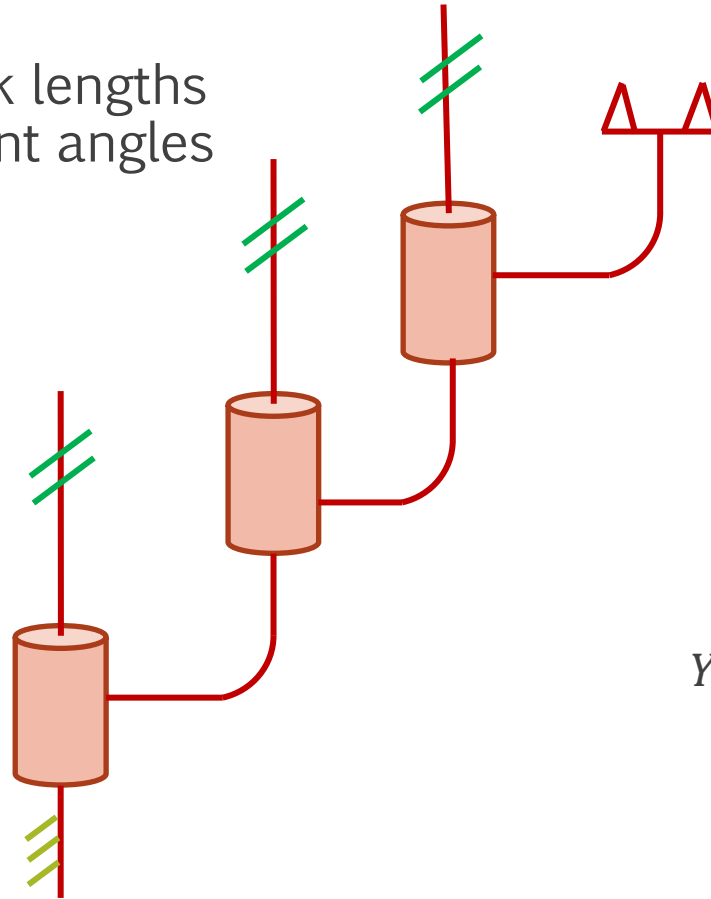
Forward Kinematics

- All three joints are revolute.
- Double hash marks indicates all 3 axes are parallel.

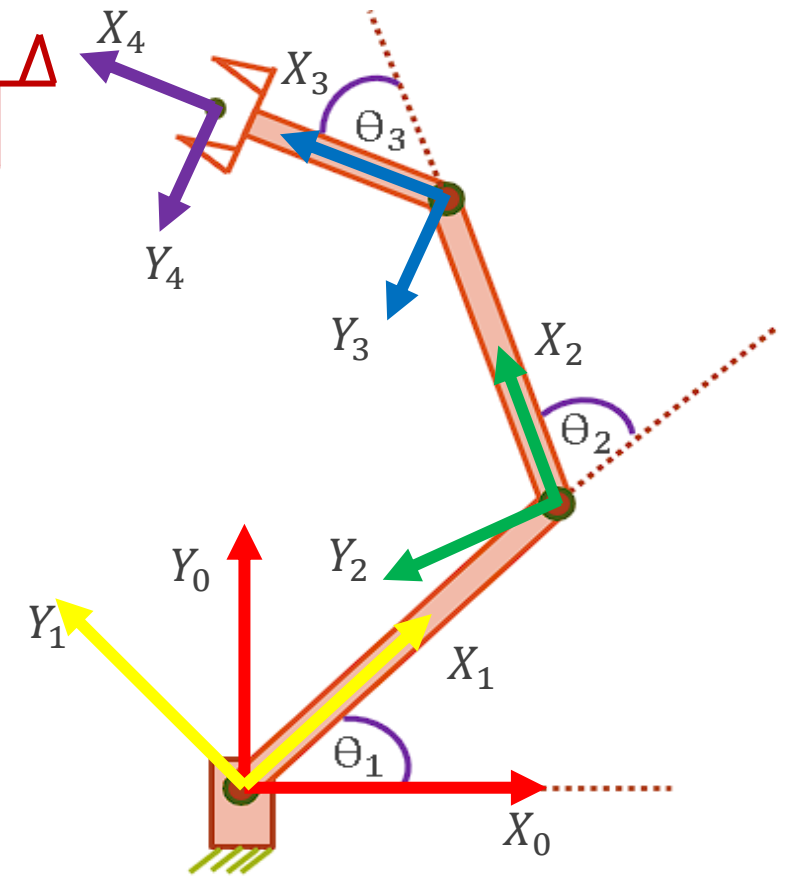


CAD Model

- $l_i \rightarrow$ Link lengths
- $\theta_i \rightarrow$ Joint angles
- $i = 1, 2, 3$



Schematic Diagram

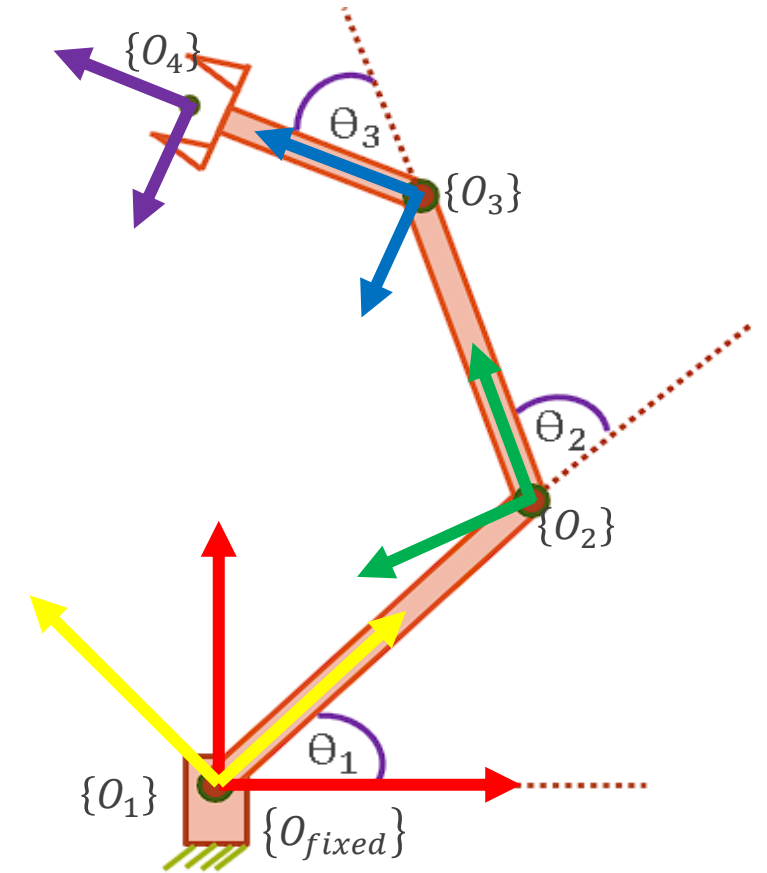


Assigning Frames

DH Parameters

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	l_1	0	θ_2
3	0	l_2	0	θ_3
4	0	l_3	0	0

- $\{O_{fixed}\} \rightarrow$ Reference frame
- $\{O_1\}, \{O_2\}$, $\{O_3\} \rightarrow$ Intermediate frames
- $\{O_4\} \rightarrow$ Tip or End-Effector frame
- $\alpha_{i-1} \rightarrow$ Link twists
- $a_{i-1} \rightarrow$ Link lengths
- $d_i \rightarrow$ Link offsets
- $\theta_i \rightarrow$ Joint angles



Link Frame Assignment

Transformation of frames

$$\square {}^{i-1}_i[T] = \text{Screw}_X(\mathbf{a}_{i-1}, \alpha_{i-1}) \text{Screw}_Z(\mathbf{d}_i, \theta_i)$$

$$= \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & a_{i-1} \\ \sin \theta_i \cos \alpha_{i-1} & \cos \theta_i \cos \alpha_{i-1} & -\sin \alpha_{i-1} & -\sin \alpha_{i-1} d_i \\ \sin \theta_i \sin \alpha_{i-1} & \cos \theta_i \sin \alpha_{i-1} & \cos \alpha_{i-1} & \cos \alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\square {}^0_4[T] = {}^0_1[T] {}^1_2[T] {}^2_3[T] {}^3_4[T]$$

$$= \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 & L_1 \\ \sin \theta_2 & \cos \theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta_3 & -\sin \theta_3 & 0 & L_2 \\ \sin \theta_3 & \cos \theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \left[\begin{array}{c|c} {}^0_4[R]_{3 \times 3} & \overrightarrow{{}^0_4p}_{3 \times 1} \\ \hline \overrightarrow{0}_{3 \times 1} & 1 \end{array} \right]_{4 \times 4}$$

- ${}^0_4[R]_{3 \times 3}$, represents orientation of tip w.r.t. fixed base
- $\overrightarrow{{}^0_4p}_{3 \times 1}$, represents the position of tip/end effector w.r.t. fixed base

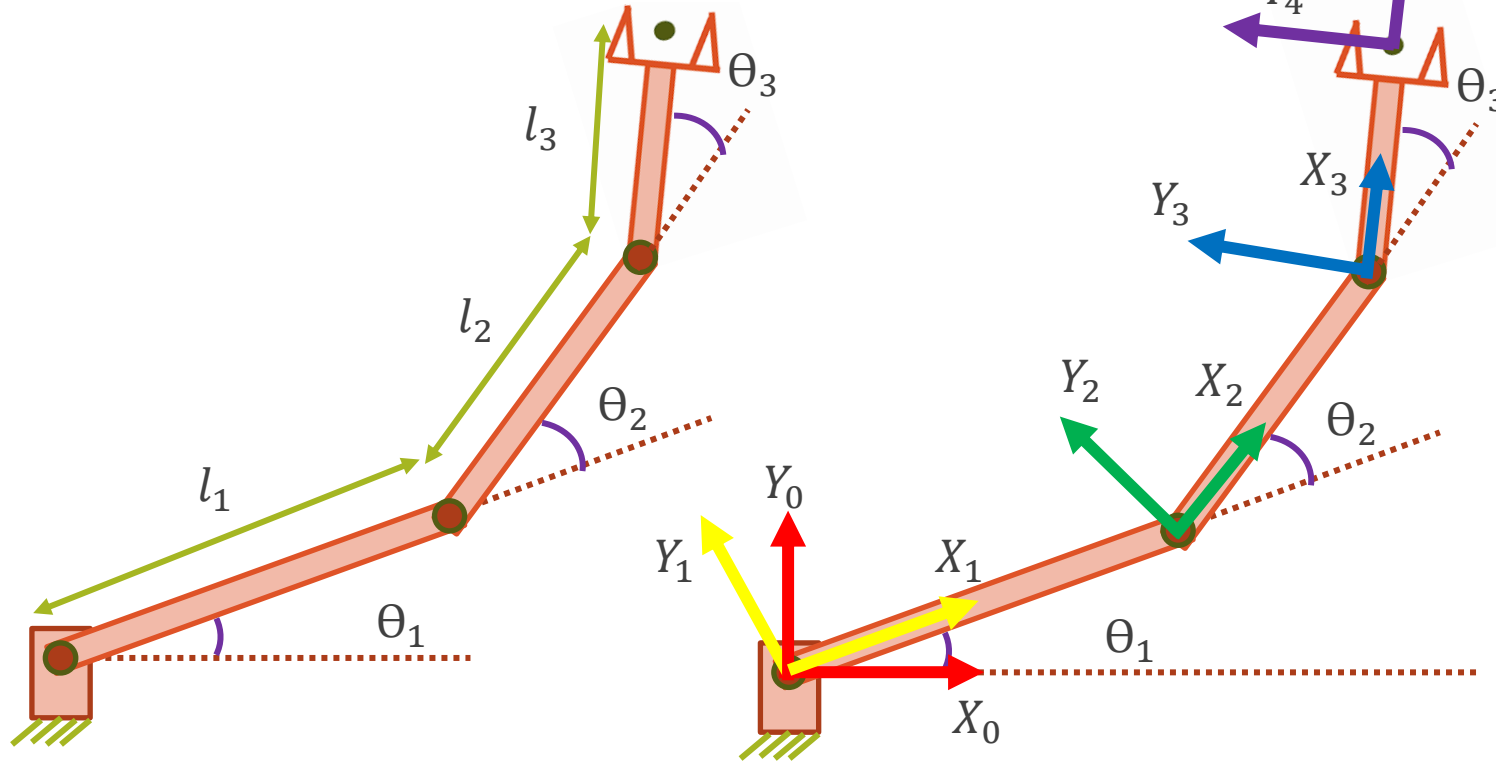
Inverse Kinematics

□ Position and orientation of the of the end-effector w.r.t. fixed base.

- $x = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2) + l_3 \cos(\theta_1 + \theta_2 + \theta_3)$

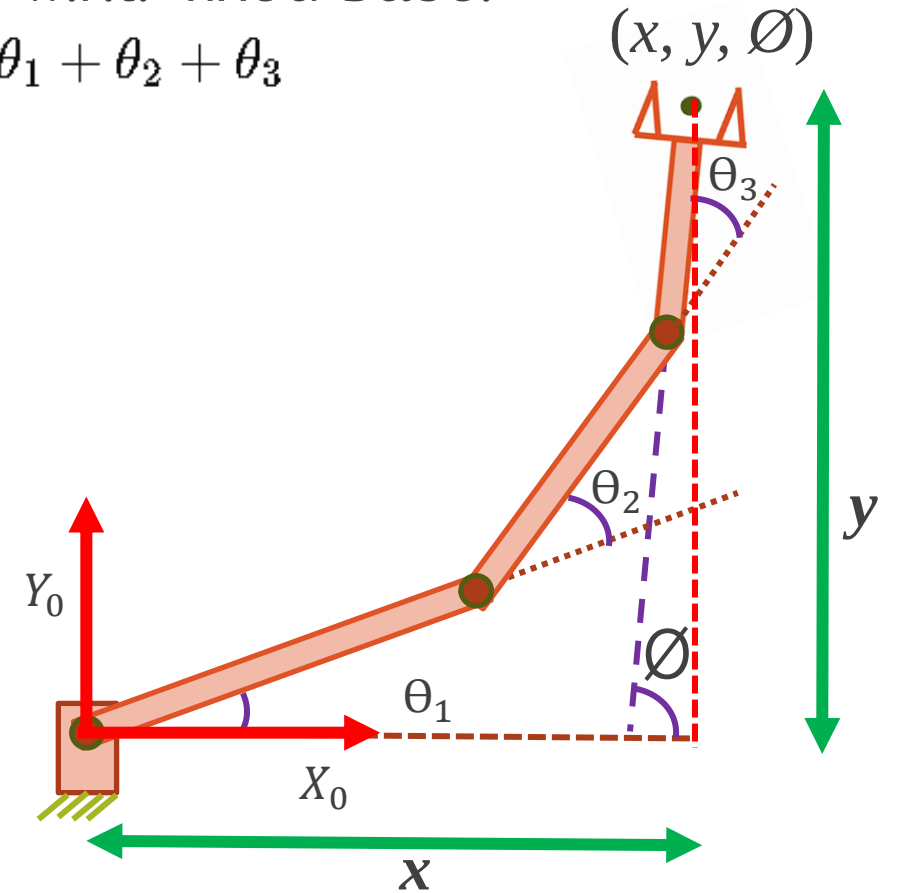
- $y = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2) + l_3 \sin(\theta_1 + \theta_2 + \theta_3)$

- $\phi = \theta_1 + \theta_2 + \theta_3$



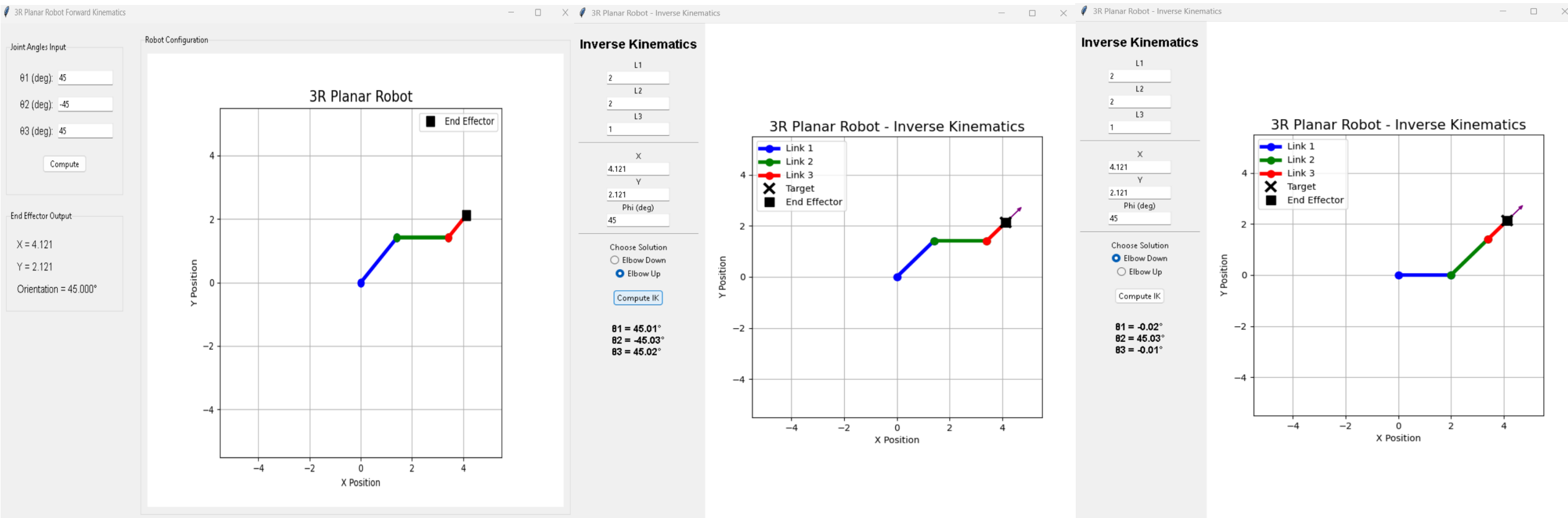
Kinematic diagram

Link Frame Assignment

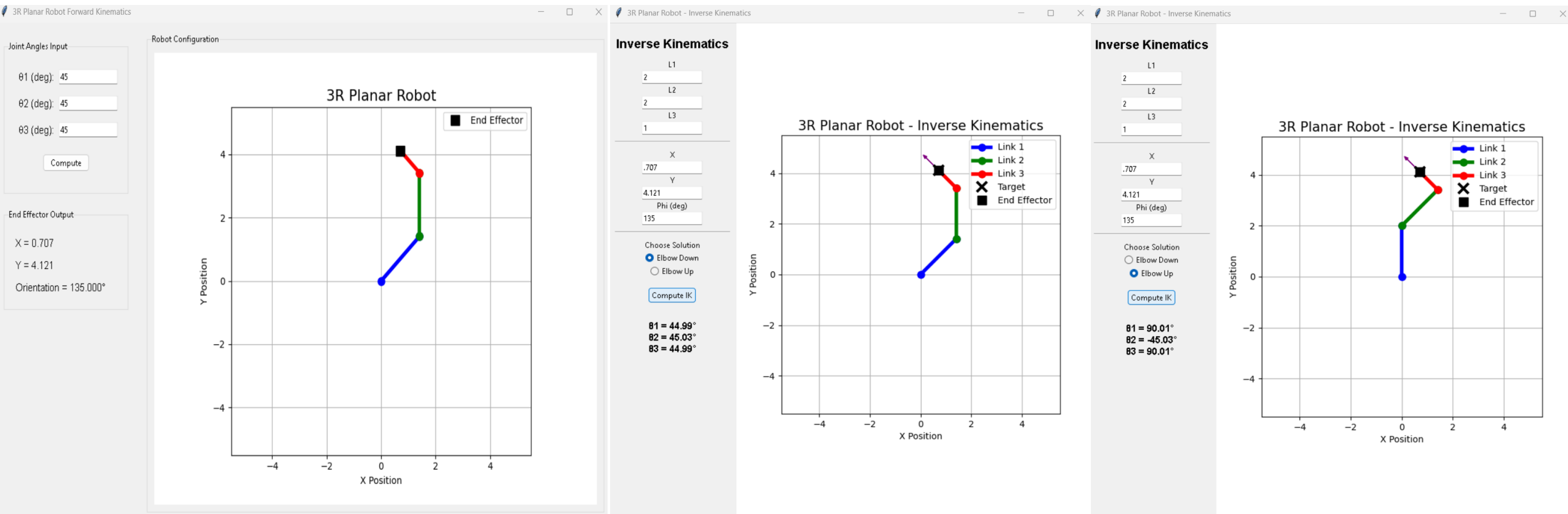


Projection of tip/end-effector's point

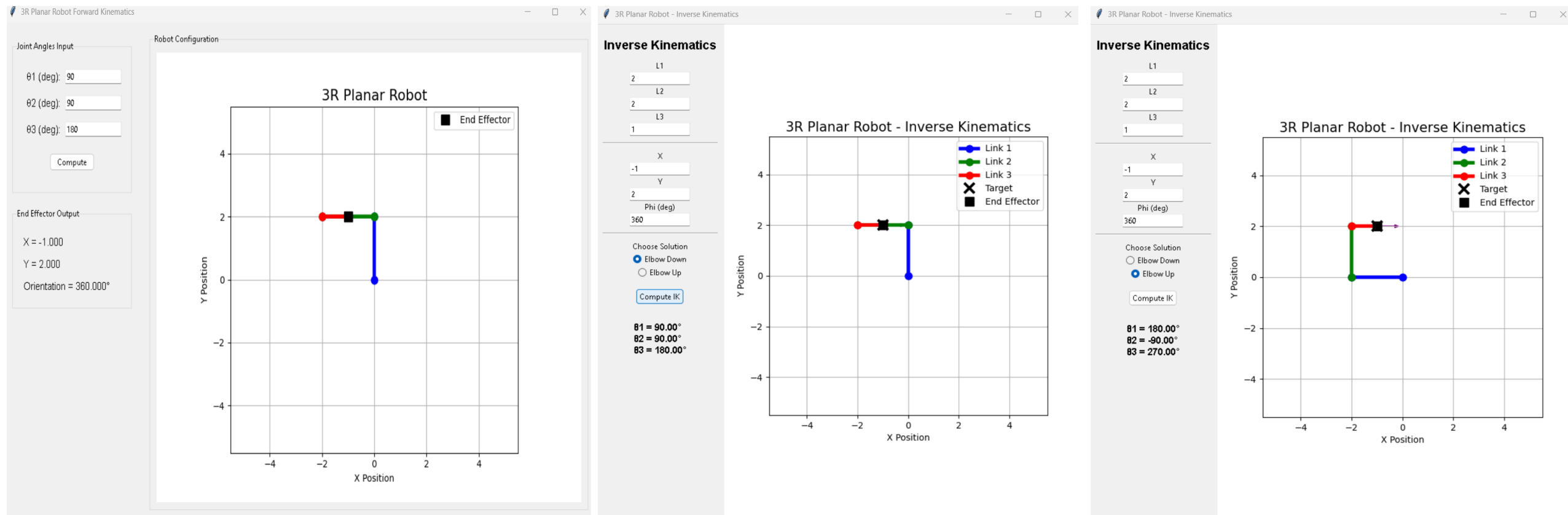
Verification of Kinematics



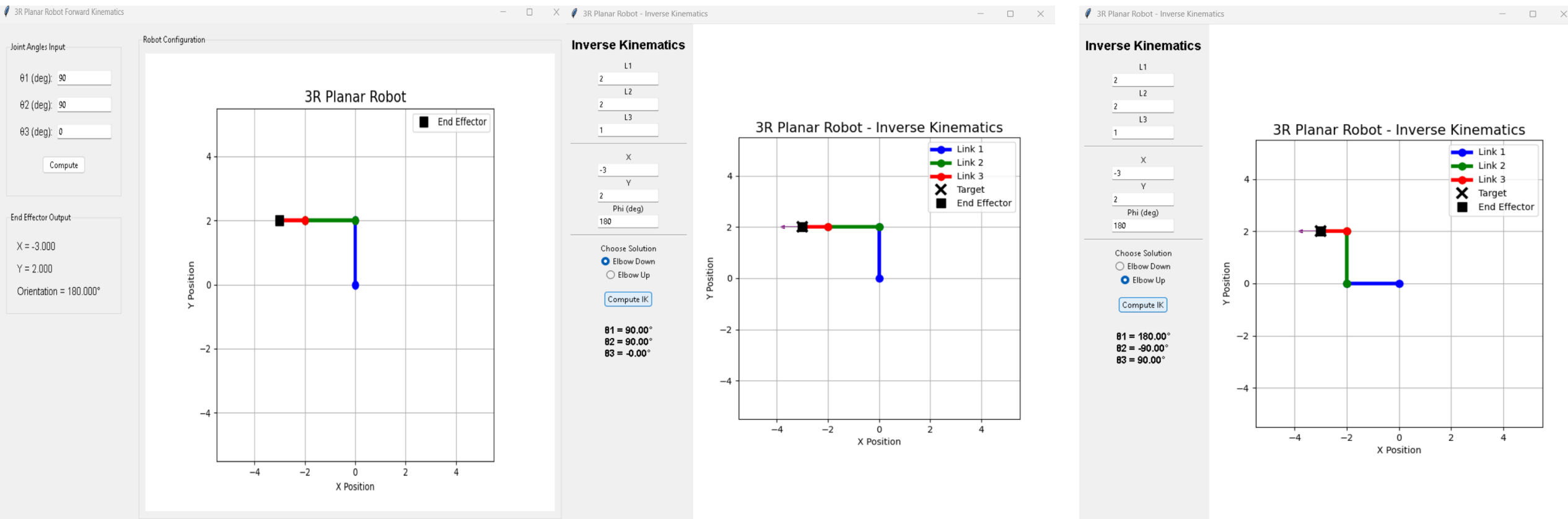
Verification of Kinematics



Verification of Kinematics



Verification of Kinematics



Thank You